

Water Potability & Safety Risk Analyzer

A machine learning system that predicts drinking water safety and explains exactly which parameters are driving the risk — turning raw lab readings into a clear, actionable recommendation.

DOMAIN
Water Safety & Public Health

DATASET
Kaggle · Water Quality & Potability (3,276 samples)

BEST MODEL
Random Forest Classifier

Problem Statement

Unsafe drinking water causes serious health issues, yet manually testing and interpreting every water quality parameter is slow and requires expertise. This project analyzes 9 lab measurements — pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity — to predict whether a sample is **potable**, and explains *why* in plain English so a non-expert can act on it immediately.

Project Workflow



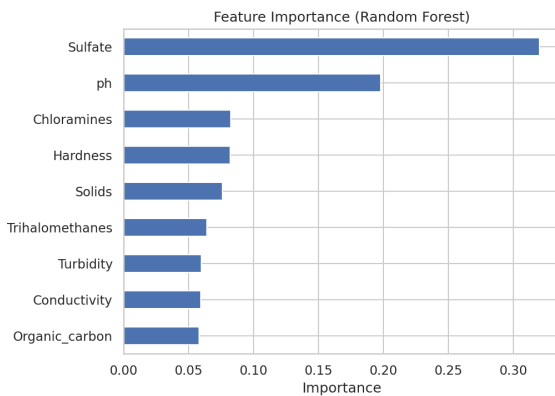
79.9%
TEST ACCURACY

0.88
ROC-AUC SCORE

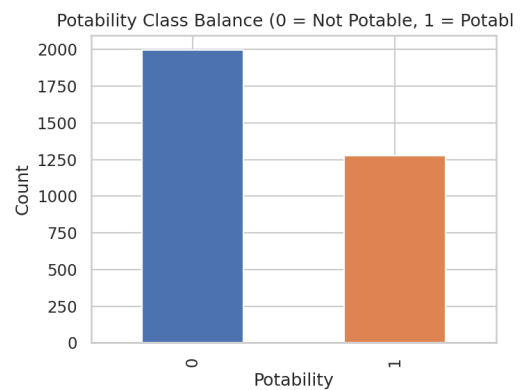
9
WATER PARAMETERS

3
MODELS COMPARED

Key Visual Insights



Feature importance — which parameters the model relies on most



Class balance — potable vs. non-potable samples in the dataset

Model Results, AI Approach & Responsible Use

How the system was built, evaluated, and where it should — and shouldn't — be trusted

Model Comparison

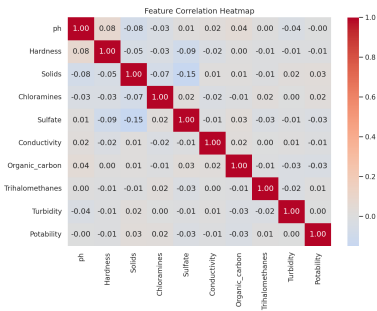
MODEL	ACCURACY	PRECISION	RECALL	F1-SCORE	ROC-AUC
Logistic Regression	61.0%	0.00	0.00	0.00	0.55
Decision Tree	75.8%	0.75	0.57	0.65	0.82
Random Forest ✓ Selected	79.9%	0.82	0.63	0.71	0.88

Insight: Logistic Regression essentially failed on this dataset — it could not draw a straight-line boundary between safe and unsafe water. This confirms potability depends on **non-linear combinations** of parameters, which only tree-based models like Random Forest capture well.

Where AI/ML Is Used

Random Forest classifier — 300 decision trees learn non-linear patterns from historical data to predict potability, with confidence scores via `predict_proba`.

Feature importance — the trained model reveals which of the 9 parameters most influence its decisions, guiding where testing effort should focus.



Correlation heatmap — confirms weak pairwise correlation, motivating a non-linear model

Tech Stack

Python

Pandas

NumPy

Scikit-learn

Matplotlib / Seaborn

Streamlit

Jupyter

joblib

Limitations & Future Work

- ! Single public dataset; may not generalize to all real-world sources

! Reference safe-ranges are simplified, not official regulatory limits

! ~80% accuracy means the tool must not be the sole safety decision-maker
- ▶ Add gradient boosting (XGBoost) and hyperparameter tuning

▶ Per-sample SHAP explainability instead of global importance only

▶ Region-specific standards (e.g. BIS 10500 for India)